

Myungkyu Lee

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RESEARCH INTERESTS

Symmetric-Key Cryptography, Cryptanalysis, Geometric Approach, Quasidifferential Framework

EDUCATION

- **Korea University, School of Cybersecurity** Mar. 2025 – Expected Feb. 2027
M.S. in Information Security, Cryptographic Algorithm Laboratory (Advisor: Prof. Seokhee Hong)
◦ GPA: 4.50/4.50 (100/100; coursework completed) Seoul, Korea
- **University of Seoul** Mar. 2020 – Feb. 2025
B.S. in Mathematics; Double Major in Computer Science
◦ GPA: 4.38/4.50 (98.8/100), Rank: 2/29 Seoul, Korea

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, E=EPRINT PREPRINT, T=THESIS, D=DOMESTIC (KOREAN)

- [E.1] **Myungkyu Lee**, Byoungjin Seok, Dongjae Lee, Deukjo Hong, Jaechul Sung, Seokhie Hong (2026). **Rotational-Quasidifferential Framework – A Geometric Approach to Rotational-XOR Cryptanalysis**. *Cryptology ePrint Archive*, Paper 2026/1633.
- [D.5] Yunjae Hwang, **Myungkyu Lee**, Insung Kim, Hanbeom Shin, Sunyeop Kim, Byoungjin Seok, Seokhie Hong, Sangjin Lee (2025). **Integral Cryptanalysis on SKINNY-64-64 using PCA**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2025 (CISC-W '25)*, Nov. 2025.
- [D.4] **Myungkyu Lee**, Yunjae Hwang, Hanbeom Shin, Insung Kim, Sunyeop Kim, Byoungjin Seok, Dongjae Lee, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Security Analysis of Alzette in the Fixed-Key Model**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2025 (CISC-W '25)*, Nov. 2025.
- [D.3] **Myungkyu Lee**, Hanbeom Shin, Insung Kim, Sunyeop Kim, Donggeun Kwon, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Hash Table Method for Data Search in Differential Cryptanalysis**. *Journal of The Korea Institute of Information Security and Cryptology (JKIISC)*, Dec. 2025.
- [D.2] **Myungkyu Lee**, Hanbeom Shin, Insung Kim, Sunyeop Kim, Donggeun Kwon, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Efficient Pair Search Techniques for Zero Difference-Based AES Cryptanalysis**. In *Proceedings of the Conference on Information Security and Cryptography Summer 2025 (CISC-S '25)*, Jun. 2025.
- [D.1] **Myungkyu Lee**, Jaechul Sung, Seokhie Hong (2024). **Randomness Testing on Block Cipher GIFT-128**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2024 (CISC-W '24)*, Nov. 2024.

AWARDS AND HONORS

- **Grand Prize (1st prize)** Fall 2025
National Cryptography Expert Training (11th Cohort), NSR & Korea Cryptography Forum
- **Special Award** Fall 2025
Cryptographic Analysis Contest, Korea Institute of Information Security and Cryptology (KIISC)
- **Grand Prize (1st prize)** Fall 2024
University Algorithm Competition (UOSPC) (Division 1), University of Seoul
- **Silver Medal (Field 1)** Fall 2024
Undergraduate Mathematics Competition, Korean Mathematical Society (KMS)
- **Special Award** Fall 2024
Cryptographic Analysis Contest, Korea Institute of Information Security and Cryptology (KIISC)
- **Grand Prize (1st prize)** Summer 2024
Gyeonggi University of Science and Technology Summer Big Data Coding Competition, Gyeonggi Univ. of Science & Tech.
- **AI Convergence Innovation Division Director's Award (3rd Prize)** Fall 2023
Industry-Academia Deep Learning Winter Bootcamp Competition, University of Seoul
- **Participation Prize (3rd prize)** Fall 2023
AICOSS Hackathon, University of Seoul
- **Participation Prize (3rd prize)** Fall 2023
University Algorithm Competition (UOSPC) (Division 2), University of Seoul
- **Participation Prize (3rd prize)** Fall 2023
Creative Engineering Mathematics Competition, University of Seoul
- **Participation Prize (3rd prize)** Summer 2023
Gyeonggi University of Science and Technology Summer Big Data Coding Competition, Gyeonggi Univ. of Science & Tech.

- **Grand Prize (1st prize)** Fall 2021
University Algorithm Competition (UOSPC) (Division 2), University of Seoul
- **Excellence Award (2nd prize)** Fall 2021
Creative Engineering Mathematics Competition, University of Seoul
- **Excellence Award (2nd prize)** Fall 2020
Creative Engineering Mathematics Competition, University of Seoul

SCHOLARSHIPS

- **Scholarship under the University Innovation Support Program** Fall 2024
University of Seoul (Division of Academic Affairs)
- **National Scholarship for Science and Engineering** Fall 2024
Ministry of Science and ICT, Republic of Korea
- **National Scholarship for Science and Engineering** Spring 2024
Ministry of Science and ICT, Republic of Korea
- **Extracurricular Education Scholarship** Spring 2024
University of Seoul
- **Scholarship under the Innovation Sharing University Project** Fall 2023
University of Seoul (Sponsored by UOS Industry Cooperation Foundation)
- **Scholarship for Excellent Achievement** Spring 2022
University of Seoul
- **Scholarship for Excellent Achievement** Fall 2021
University of Seoul
- **Seoul Mayor's Scholarship (Top Honor)** Spring 2021
University of Seoul
- **Scholarship for Excellent Achievement** Fall 2020
University of Seoul

TEACHING AND MENTORING EXPERIENCE

- **SHA (UOS Hacking Club) — Cryptography Study Group** Fall 2024
Study Leader; Mentor
- **Elementary Linear Algebra** Fall 2024
Teaching Assistant (Paid Position, Lecturer: Prof. Jeong-Ah Kim)
- **Mathematics and Computer** Fall 2024
Teaching Assistant (Paid Position, Lecturer: Prof. Jaechul Sung)
- **PseudoCode (Algorithm Club in the Department of Mathematics, UOS)** Summer – Winter 2024
Founder & Club Leader
- **Discrete Mathematics** Spring 2024
Teaching Assistant (Paid Position, Lecturer: Prof. Jaechul Sung)
- **Calculus II** Fall 2021
Course Tutor (Paid Position)

ADDITIONAL ACADEMIC TRAINING

- **National Cryptography Expert Training (11th Cohort)** May – September 2025
Selected Trainee (15 selected nationwide); Organized by NSR & Korea Cryptography Forum

MILITARY SERVICE

- **Republic of Korea Army** Jan. 2022 – Jul. 2023
Compulsory Military Service as a C4I Systems Operator and Maintainer

TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++ (Advanced), SageMath (Basic)
- **Frameworks/Tools:** PyTorch, TensorFlow, Flask (Intermediate)

ADDITIONAL INFORMATION

Languages: Korean (native), English (advanced)
Interests: Squash, Badminton, and Calisthenics